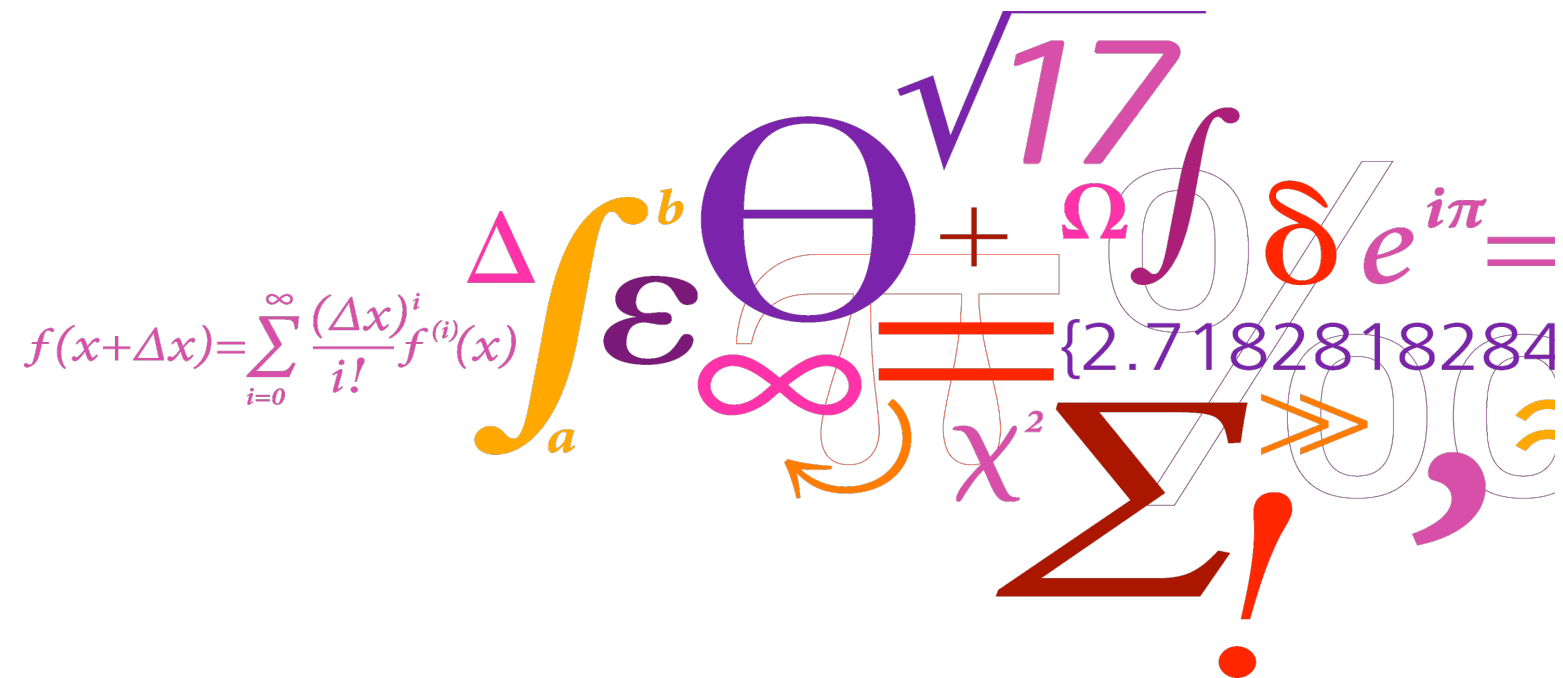


Research Topics in Cybersecurity (02234)

Fall 2026


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Nicola Dragoni

Professor in Secure Pervasive Computing, PhD

Deputy Director @ DTU Compute

Head of Section Cybersecurity Engineering

 ndra@dtu.dk

DTU Compute

Department of Applied Mathematics and Computer Science

Technical University of Denmark (DTU)

Richard Petersens Plads

Building 322, room 217

2800 Kongens Lyngby

Today's Agenda

1. Course Practicalities

- Course format
- Activity plan
- Exam
- Example of Technical Paper (Alfonso)

2. From Logs to Cyber Threat Intelligence: Mining the Behaviour of Cyber Threats

Communication: How and When

- **Course structure and organization (including student workshop):**
 - Send me an email: ndra@dtu.dk
- **Specific lecture in the activity plan:**
 - Send an email to the corresponding lecturer
- **N.B.:** Always include **[02234]** in the **SUBJECT** of your email
- If **time-consuming issue**, **ask for a meeting** instead of writing a loooooooooooooong email
 - With me, concerning the overall structure and organisation of the course
 - With the lecturer, concerning the content of a specific lecture
- **In person**, during breaks or after each lecture

Teaching Assistant



s224818@student.dtu.dk



Melissa Safari  · 1st

Stud. MSc in Engineering - Computer Science - @ DTU |
Cybersecurity |

Copenhagen, Capital Region of Denmark, Denmark · [Contact info](#)



SJF Bank



DTU - Technical University of
Denmark

Course Web Pages

<https://kurser.dtu.dk/course/02234>

Academic prerequisites 02270.(02271/02232) , The course is an advanced course that requires a solid understanding of computer and network security.

<http://www2.imm.dtu.dk/courses/02234/>
(activity plan)

Course Format

- Combination of lectures and project work
- Lectures:
 - **11 research subjects**, related to cybersecurity and cryptography research at DTU Compute, will be covered in lectures 1 - 11
 - Students should expect to read some scientific papers in preparation for each lecture or as follow up of the lecture
 - Lectures cover different advanced subjects in cybersecurity and cryptography
 - Lectures start at 1pm and last for 1-2 hours (depending on the specific lecture)
 - Remaining time is for project work (including reading material)
 - Each lecture features a different researcher/teacher!

Activity Plan

[Also here](#)

Please note that the activity calendar below is preliminary and might be subject to change.

Lecture N.	Date	Activity	Lecturer	Topic
1	01 September	Lecture	Nicola Dragoni Alfonso Pedro Ridao	Course presentation (Nicola) Example of Technical Paper (Alfonso): <i>MCP-SecLint: An Open-Source Static Analyzer for Detecting Vulnerabilities in LLM Tool Integrations</i> A. Ridao, M. Safari, Z. Kang, N. Dragoni <i>In Proceedings of the 12th ACM International Workshop on Security and Privacy Analytics (IWSPA), 2026.</i> From Logs to Cyber Threat Intelligence: Mining the Behaviour of Cyber Threats (Nicola)
2	08 September	Lecture	Alberto Maria Mongardini Nicola Dragoni	Measuring Online Harm at Scale across Blockchain and Online Social Network Ecosystems (Alberto) How to write a scientific paper (Nicola)
3	15 September	Lecture	Sebastian Alexander Mödersheim	Logic for Security
4	22 September	Lecture	Gaurav Choudhary	Next Generation Technology-Driven Digital Era: Cyber Security and Ethics in the CPS-IoT Space How to write a systematic literature review
5	29 September	Lecture	Lejla Islami	Conducting Research in Usable Privacy and Security: It's Complicated!
6	06 October	Lecture	Edlira Dushku	Remote Attestation for IoT
Autumn Break				
7	20 October	Lecture	Luisa Siniscalchi	Exploring the Beautiful World of Secure Multi-Party Computation
8	27 October	Lecture	Emmanouil Vasilomanolakis	An Introduction to Cyber-Deception
9	03 November	Lecture	Christian Majenz	The Signal Protocol for Instant Messaging Key Exchange
10	10 November	Lecture	Carsten Baum	Privacy in Practice: The EU Digital Identity Wallet
11	17 November	Lecture	Tyge Tiessen	Kerckhoffs's Principle and Standardization in Cryptography
12	24 November	Student Workshop Feedback on Projects	Nicola Dragoni	Schedule will be published on DTU Learn
13	01 December	Student Workshop Feedback on Projects	Nicola Dragoni	Schedule will be published on DTU Learn

N.B.

- It is recommended to attend the lectures
 - To get to know hot topics in cybersecurity and cryptography
 - To select the best project for you and your group
 - To get to know (top) cybersecurity researchers at DTU Compute
 - To get inspired and start planning your **MSc thesis project**
 - Start as early as possible!

(No) Streaming of Lectures

- The default form of lectures is **physical**
 - I.e., no hybrid, no streaming
- It is the individual lecturer who decides whether to stream the lecture or not, changing the form of the lecture to hybrid or fully online
 - This will be communicated in advance on DTU Learn
- In case nothing is communicated on DTU Learn, you should **not** expect the lecture to be streamed (i.e., default is in person)

What You Have to Do in This Course

(I) **Select one topic** for the **group project work**

- Might be one of the topics covered in the lectures

OR

- any other subject relevant to cybersecurity

N.B.: Subject has to be first approved by me if not part of any lecture!

(II) **Present** the research project during the Student Workshop

(III) **Write** a research paper as outcome of the group project

(IV) **Submit** the paper on DTU Learn before **midnight (CET) on January 3rd, 2027**

Groups

- Groups of **2-3 students**
 - This means **NO solo** projects, **NO groups of > 3 students**
 - **Self-organisation!**
 - A **forum** will be created on DTU Learn to facilitate this
 - TA will send announcement when forum is ready
 - **Registration** of the group on DTU Learn
 - TA will send an announcement with related instructions

How Many of You Have Written / Published
a **Research Paper**?

Report (Research Paper)

- Two types of reports:

1. Technical paper

- solve a cybersecurity problem by **designing** (and optionally implementing) an appropriate solution (framework/protocol/system/...)

2. Systematic literature review

- analyse a cybersecurity problem by **gathering, reviewing, comparing** and **classifying** all relevant literature

Type of Report: **Technical Paper**

- Target a security problem
- Propose a **new solution** to that problem
 - This represents your contribution!
- Briefly analyse existing literature (solutions and approaches) to motivate why a new solution is needed
- Relate the new solution to existing literature (**novelty** of new proposal)

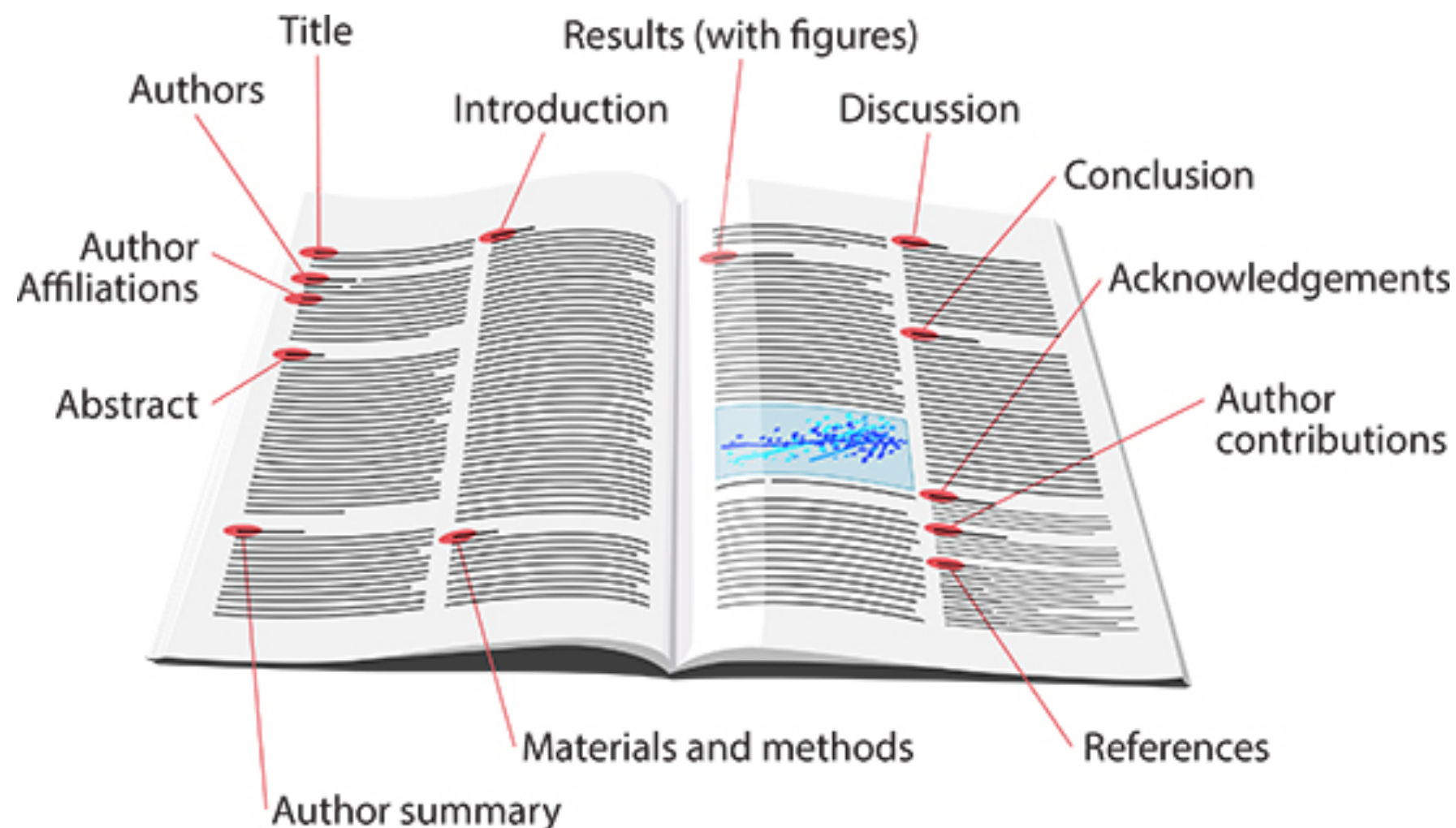


Examples of Technical Papers

- **Bruteware: A Novel Family of Cryptoviral Attacks.** V. Coletti, E. Müller, F. Meyer Lehmann Lundsberg, N. Dragoni. In Proceedings of the 39th International Conference on ICT Systems Security and Privacy Protection (IFIP SEC), 2024, Springer.
- **ARCADIS: Asynchronous Remote Control-Flow Attestation of Distributed IoT Services.** R. M. Halldórsson, E. Dushku, N. Dragoni. In IEEE Access, vol. 9, 2021.
- **PERMANENT: Publicly Verifiable Remote Attestation for Internet of Things through Blockchain.** S. Ankergård, E. Dushku, N. Dragoni. In Proceedings of the 14th International Symposium on Foundations & Practice of Security (FPS), 2021, Springer.
- **ERAMO: Effective Remote Attestation through Memory Offloading.** J. H. Østergaard, E. Dushku, N. Dragoni. In Proceedings of the IEEE International Conference on Cyber Security and Resilience (CSR), 2021, IEEE.
- **rTLS: Lightweight TLS Session Resumption for Constrained IoT Devices.** K. Tange, D. Howard, T. Shanahan, S. Pepe, X. Fafoutis, N. Dragoni. In Proceedings of the International Conference on Information and Communications Security (ICICS), 2020, LNCS.
- **Evaluation of Professional Cloud Password Management Tools.** D. Schougaard, N. Dragoni, A. Spognardi. In Proceedings of the 2nd International Workshop on TEchnical and LEgal aspects of data pRivacy and SEcurity (TELERISE'16), affiliated with ICWE'16, 2016, LNCS.
- **Adding Salt to Pepper - A Structured Security Assessment over a Humanoid Robot.** A. Giaretta, M. De Donno, N. Dragoni. In Proceedings of the 13th International Conference on Availability, Reliability and Security (ARES'18), IoT-SECFOR session, 2018, ACM.
- **SxC4IoT: A Security-by-Contract Framework for Dynamic Evolving IoT Devices.** A. Giaretta, N. Dragoni, F. Massacci. In ACM Transactions on Sensor Networks, 18(1): 12, 1-51, 2021.
- **SHAKE: SHared Acceleration Key Establishment for Resource-Constrained IoT Devices.** E. Bejder, A. K. Mathiasen, M. De Donno, N. Dragoni, X. Fafoutis. In Proceedings of the 6th IEEE World Forum on Internet of Things (WF-IoT), 2020, IEEE.

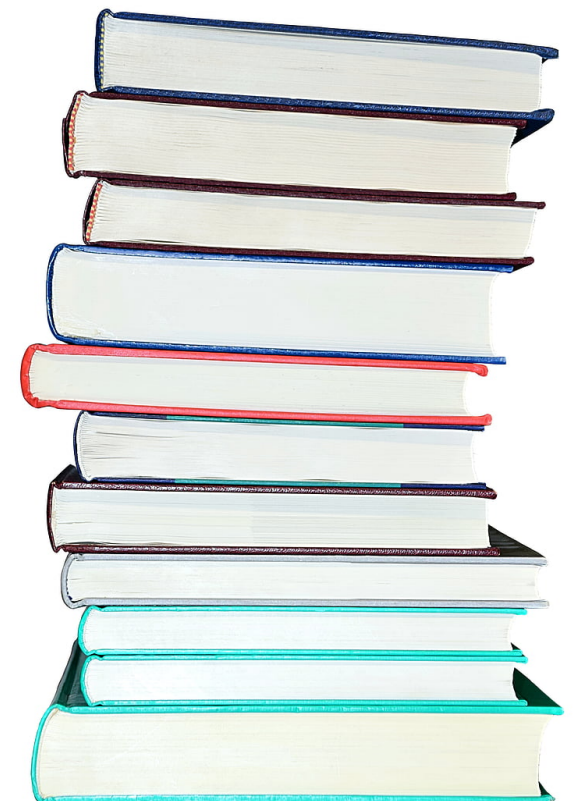
More Details in the “How to” Lecture

- Lecture: how to write a scientific technical paper (08th of September)
- Read “How to write a first-class paper” (today’s material on DTU Learn or click [here](#))



Type of Report: **Systematic Literature Review**

- Target a security problem
- Systematically review and compare existing literature (approaches and solutions) addressing that problem
 - **Contribution is the analysis and the systematisation of knowledge!**
 - Analysis has to be complete and deep!
- Discuss what is missing in the literature to solve the problem

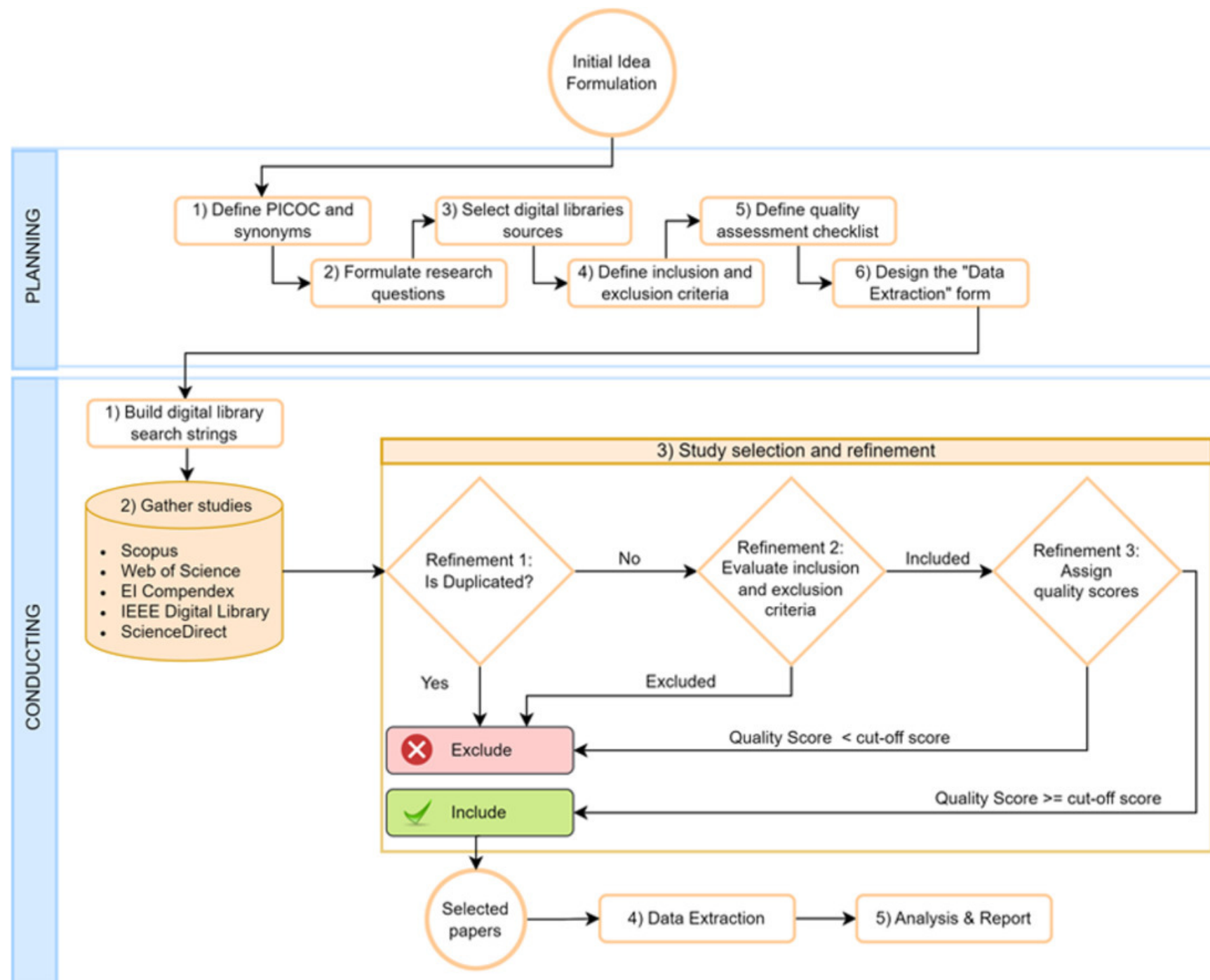


Examples of Systematic Literature Reviews

- **Systematic Literature Review of IoT Botnet DDOS Attacks and Evaluation of Detection Techniques**
M. Gelgi, Y. Guan, S. Arunachala, M. Samba Siva Rao, N. Dragoni
In Sensors, 24(11), 3571, 2024. DOI: <https://doi.org/10.3390/s24113571>.
- **Cyber4Drone: A Systematic Review of Cyber Security and Forensics in Next-Generation Drones**
V. Sihag, G. Choudhary, P. Choudhary, N. Dragoni
In Drones, 7(7): 430, 2023. DOI: <https://doi.org/10.3390/drones7070430>.
- **ML-based 5G Network Slicing Security: A Comprehensive Survey**
R. Dangi, A. Jadhav, G. Choudhary, N. Dragoni, M. K. Mishra, P. Lalwani
In Future Internet, 14(4), 116, 2022. DOI: <https://doi.org/10.3390/fi14040116>.
- **A Survey on Botnets: Incentives, Evolution, Detection and Current Trends**
S.N.T. Vu, M. Stege, P.I. El-Habr, J. Bang, N. Dragoni
In Future Internet, 13(8), 198, 2021. DOI: <https://doi.org/10.3390/fi13080198>.
- **State-of-the-Art Software-Based Remote Attestation: Opportunities and Open Issues for Internet of Things**
S. Ankergård, E. Dushku, N. Dragoni
In Sensors, 21(5), 2021.
- **A Systematic Survey of Industrial Internet of Things Security: Requirements and Fog Computing Opportunities**
K. Tange, M. De Donno, X. Fafoutis, N. Dragoni
In IEEE Communication Surveys & Tutorials, 22(4): 2489-2520, 2020.
- **Blockchain Implementations and Use Cases for Supply Chains – A Survey**
P. Gönczöl, P. Katsikouli, L. Herskind, N. Dragoni
In IEEE Access, 8, 2020.

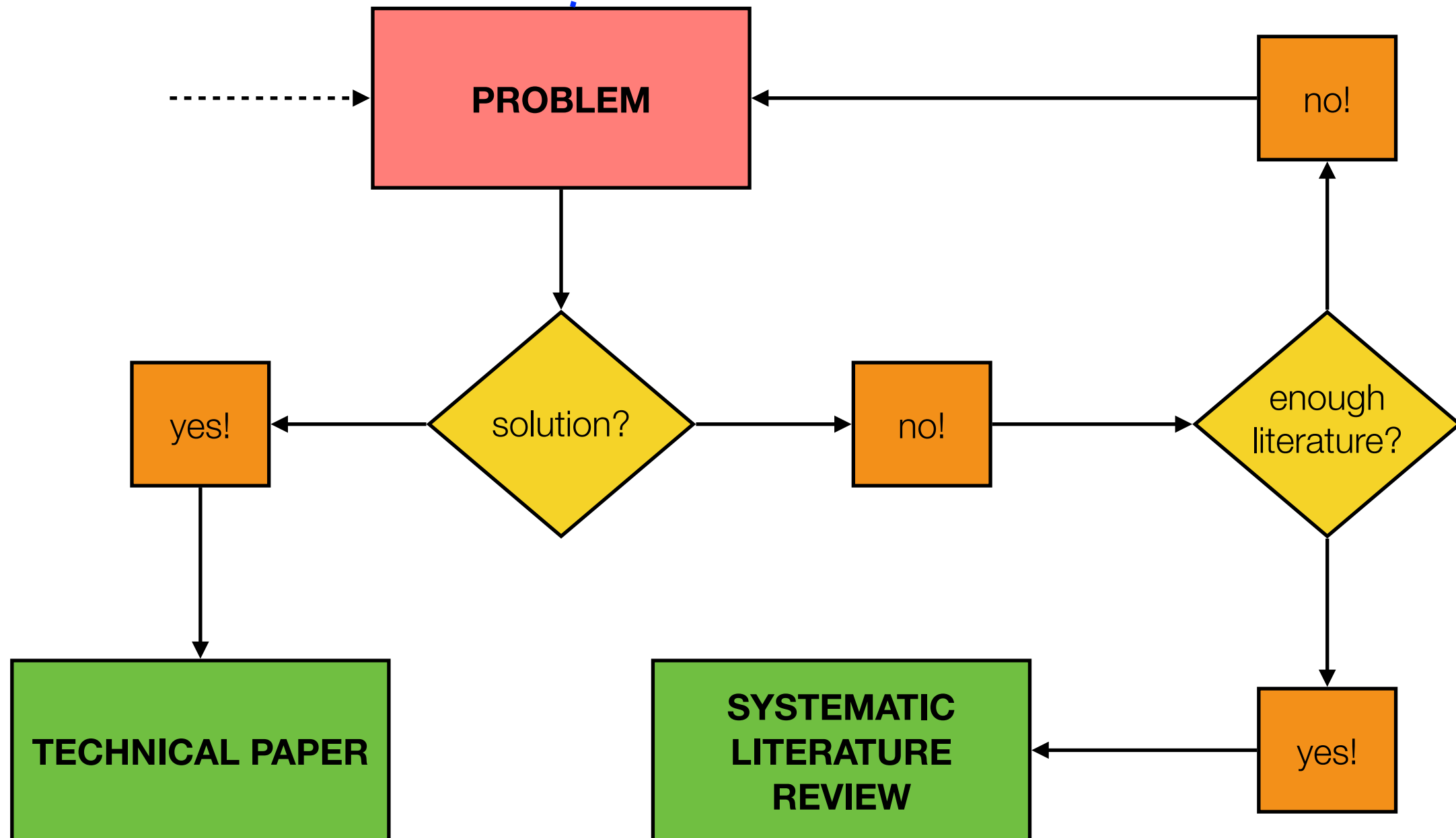
More Details in the “How to” Lecture

- Lecture: how to write a systematic literature review (22nd of September)

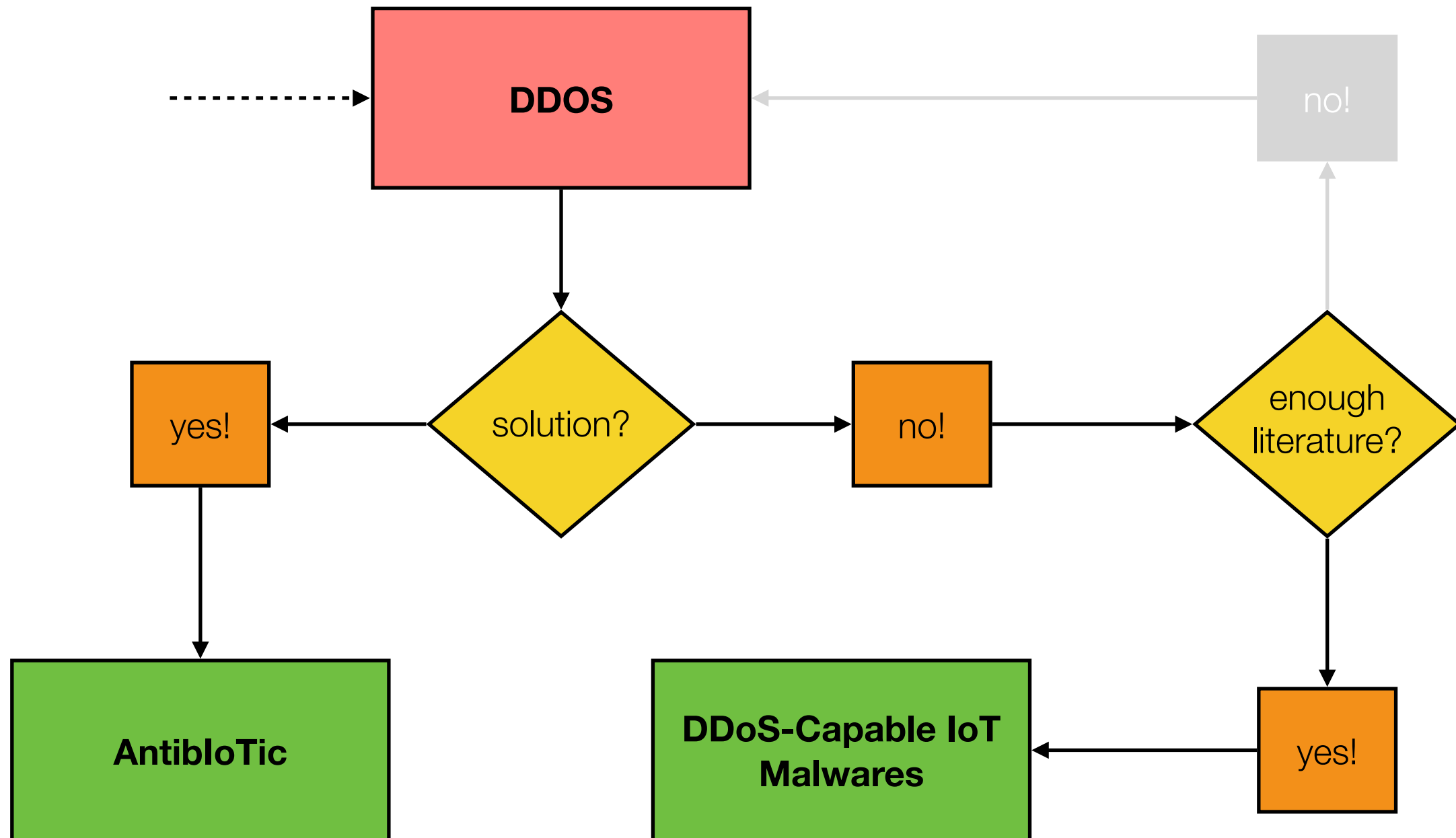


Recap

See “What is a problem statement”
(today’s material on DTU Learn)



Recap - Example



Template for Project Reports

- Lecture Notes in Computer Science by Springer:

<https://www.springer.com/gp/computer-science/lncs/conference-proceedings-guidelines>

- Available also on overleaf:

<https://www.overleaf.com/latex/templates/springer-lecture-notes-in-computer-science/kzwwpvhwnvfj>

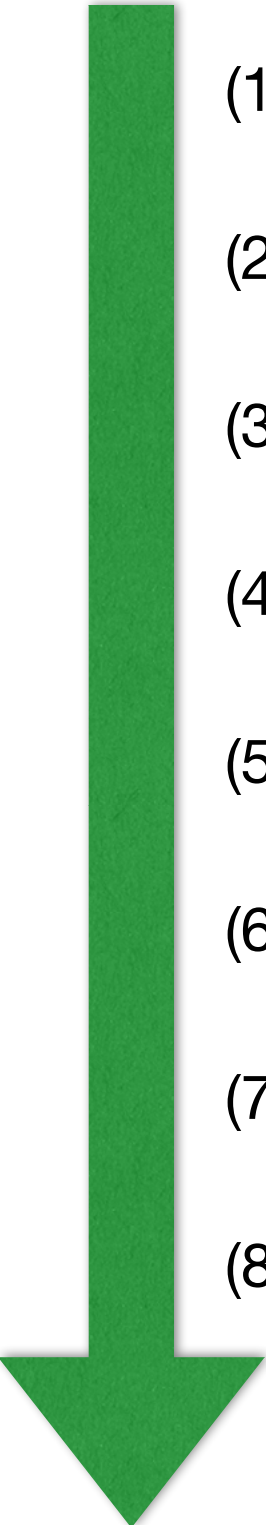
- Max **16 pages** for **technical reports** (excluding references)
- Max **30 pages** for **systematic literature reviews** (excluding references)



!!! USE THE TEMPLATE !!!



[Recap] Steps

- 
- (1) Attend lectures
 - (2) Search for project mates (use DTU Learn forum if needed)
 - (3) Brainstorm on possible project (including project type)
 - (4) Register the group on DTU Learn (required for handing-in your project)
 - (5) Work on the project
 - (6) Present the project status and progress at the student workshop
 - (7) Ask for feedback
 - (8) Hand in the final project report by the deadline (January 3rd)

How to Find a Research Subject?

- ☑ Attend lectures
- ☑ Talk to lecturers during lecture breaks and/or at the end of the lectures and/or in a planned meeting
- ☑ Read scientific papers
 - Most papers indicate **future work**
 - It's totally fine to extend an existing paper with **novel contribution**
- ☑ Read official reports on cyber threats
- ☑ If you work for a company, ask for concrete / real world problems
 - BUT your project must be framed as **research project** (problem, literature review, novel proposal)
 - Only implementation/coding is **not** enough

Cheating and AI Chatbots

DTU considers it exam cheating among other things in the following cases:

1. Plagiarism

Plagiarizing means using someone else's text, results, ideas, images, figures, slides (also from teaching), notes, data, etc. as your own without indicating the precise source. **Use of AI, such as ChatGPT and similar forms of artificial intelligence, where it is not you, but the program/robot that writes, solves a task, etc., is also plagiarism if you do not cite the source. Refining an assignment's content using grammar or spelling checks is widely accepted and does not require citation.** It must be clear from an assignment answer, etc., which parts are the result of one's own thoughts, and which are based on direct copying or processing of other people's knowledge, **including from a program/robot.** When using texts, images, graphs, etc., which are not your own, whether as a quote, paraphrase, report, translation, or assessment, a reference to the source must be made. A paraphrase is a reformulation of a text. Read more about plagiarism and correct citations and references at www.stopplagiat.nu which is a web tutorial for students about plagiarism.

Full text: [Cheating at exams and other forms of assessment](#)

DTU's Code of Honour

"As a student at DTU I acknowledge DTU's code of honour for examinations and other academic activity. I accept and respect that the high standards for accountability and academic and scientific integrity apply to me as a student just as they apply to lecturers, researchers, and students at the university.

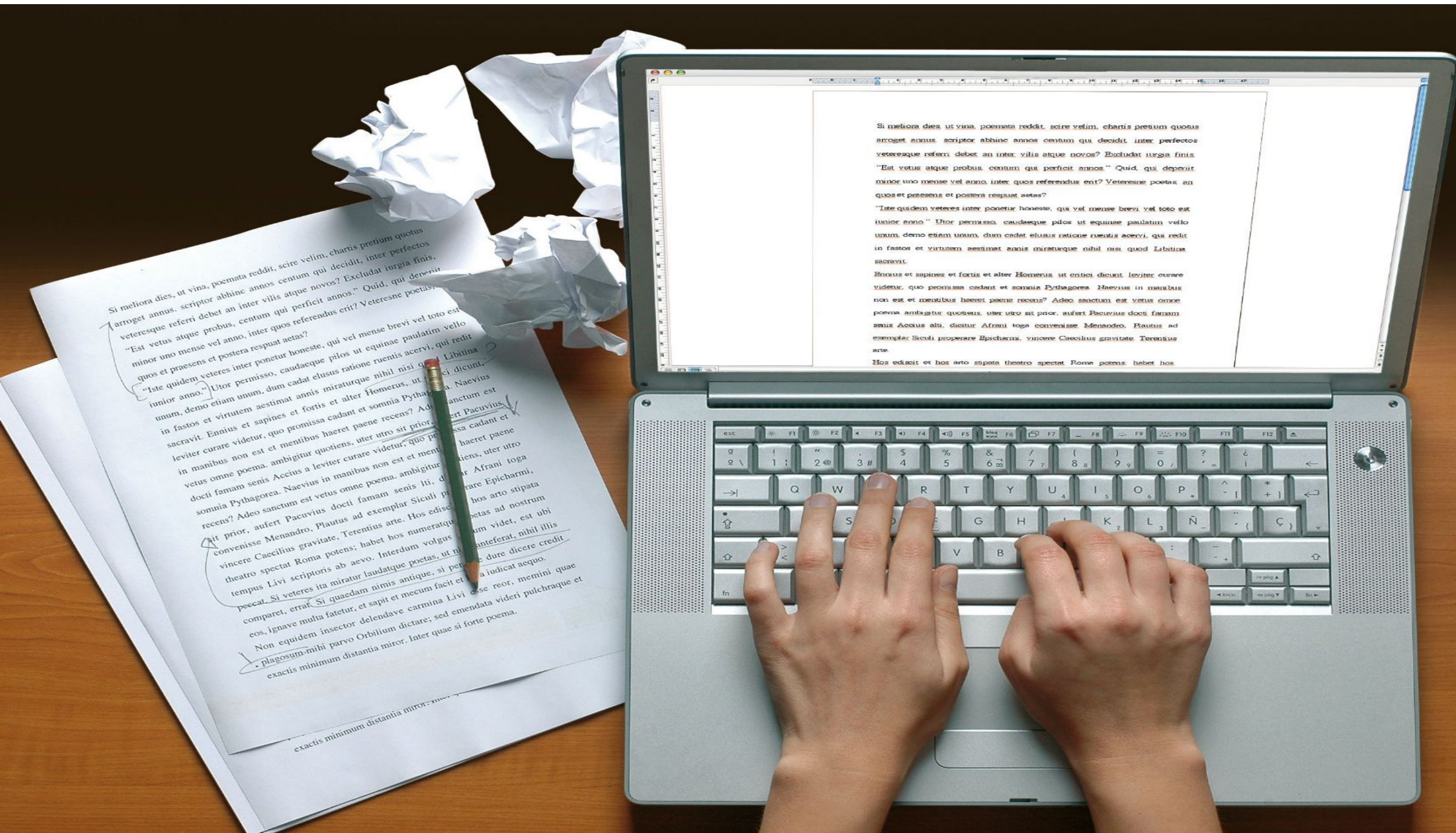
I therefore show responsibility and integrity in my work, and my exam submissions always reflect my own work, without having received inappropriate oral or written assistance, including digital sources. I know that I must never copy (plagiarize) other people's ideas, thoughts, reports or articles, but I am permitted to quote and refer to them using quotation marks and source references. I also know that I am not allowed to communicate with others during a written exam."

- [Read about exam cheating and how to avoid it](#)
- [Read about sanctions for cheating and plagiarism](#)



Publication of Your Report

Top quality reports might get invited for publication under lecturer's supervision





Melissa Safari

2mo · Edited

...

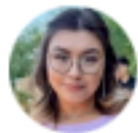
Yesterday I travelled on my own to Frankfurt, to represent our research team ([Alfonso](#), [Zachary](#), [Nicola](#), and myself) and [DTU - Technical University of Denmark](#), presenting our published paper about securing MCP servers in the development phase at The 16th ACM Conference on Data and Application Security and Privacy x IWSPA.

This will most likely be a moment I'll cherish forever. It was the first time for me standing at a podium in front of an audience, discussing our research and engaging with other researchers from all over the world.

And since the conference took place in Frankfurt, I (of course), had to make a stop by the European Central Bank.

Overall, grateful for the opportunity to represent our team.





Andreea Asimine • 1st

Student Assistant at Siemens Gamesa | MSc Computer Science ...

2mo •

I'm very happy to share that our research paper, "PromptShield: Protecting User Privacy in AI Chatbots," has been published in **Software MDPI**, an **MDPI** journal.

The paper explores how AI chatbots can better protect user privacy through a privacy-by-design framework that detects and sanitizes sensitive information before it reaches the chatbot, while also giving users more transparency and control over stored data.

This work was developed together with **Fernando Salhab Doria**, **Francisco Torgo**, **Gaurav Choudhary, Ph.D.** and **Nicola Dragoni** at the **DTU - Technical University of Denmark**.

I'm grateful to my colleagues and supervisors for the collaboration, guidance, and support throughout this research process. It was a very valuable experience to contribute to a topic that is becoming increasingly important as AI systems become part of education, work, and everyday life.

You can read the paper here: <https://lnkd.in/eu6WtWsi>



PromptShield: Protecting User Privacy in AI Chatbots



Published in Software • 2026



Technical University of
Denmark (DTU)



Andreea Isabel Asimine,
Fernando Salhab Doria Ricardo,
Francisco Torgo, Gaurav Choudhary,
Nicola Dragoni



Privacy-preserving AI chatbots



PromptShield: Protecting User Privacy in AI Chatbots

Andreea Isabel Asimine, Fernando Salhab Doria Ricardo, Francisco Torgo, Gaurav Choudhary, and Nicola Dragoni

Section of Computer Engineering, Department of Applied Mathematics and Computer Science, Technical University of Denmark (DTU), 2800 Lyngby, Denmark; andrea@dtu.dk (A.A.); fernando@dtu.dk (F.S.D.R.); francisco@dtu.dk (F.T.); gaurav@dtu.dk (G.C.); nicola@dtu.dk (N.D.)

Abstract

Artificial Intelligence (AI) chatbots are becoming important components in education, professional, and personal contexts. Users frequently disclose sensitive information without the necessary awareness of how their data is processed, stored, and later used, creating risks of unintended exposure and violation of data protection regulations such as GDPR and the EU AI Act. In this paper, we present a dynamic privacy-by-design framework that introduces a novel privacy-oriented middleware layer between the users and the AI Chatbots. Our framework intercepts user prompts, detects and sanitizes sensitive information, and enables explicit user control over data retention and disclosure through our dedicated Information Manager Dashboard. To develop our framework around real-world needs, we conducted a survey with 60 participants investigating privacy concerns, regulatory awareness, and preferences for transparency and control in conversational AI. Our results indicate that better and more transparent privacy safeguards can be achieved without significantly compromising usability or performance, supporting the development of trustworthy and user-centric AI chatbots. We also evaluate our solution against other state-of-the-art implementations and relevant metrics.

Keywords: artificial intelligence; AI chatbots; user privacy; security; GDPR; sensitive information; data protection; prompt sanitization

1. Introduction

Artificial Intelligence (AI) has reshaped the way we interact with technology across most aspects of our daily lives, in educational, professional, and everyday contexts. One of its most common uses comes in the form of AI chatbots, conversational-based systems that communicate with users via text, images or speech. These systems utilize Large Language Models (LLMs), which are advanced algorithms that are trained on extensive data, as a way to interpret context and generate responses similar to how a human would.

While AI chatbots are increasing in popularity and rapidly developing their capabilities, there is also a rise in substantial issues regarding user data privacy, control, and data handling transparency. It is common for users to frequently share their personal or sensitive information without understanding how their data is processed, stored, or reused for other purposes. Such practices introduce ethical and legal risks, namely when private or confidential information is unintentionally exposed [1]. We can take as examples students who upload personal essays or academic records to get writing assistance, patients who describe medical symptoms or medical information to receive health-related advice [2], or even

Software 2026, 5, 34

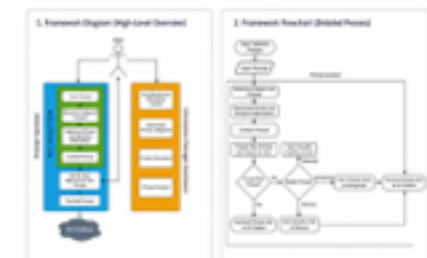
<https://doi.org/10.3390/software5050034>

1 / 3

THE PROPOSED FRAMEWORK

How PromptShield Works

A privacy-aware middleware layer between the user and the AI chatbot.



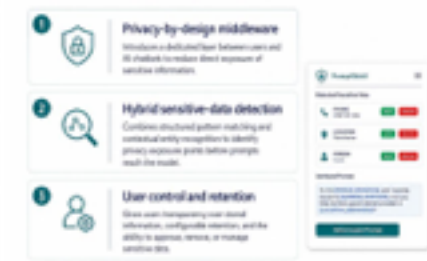
PromptShield detects sensitive information, lets users decide what to keep or remove, and forwards a safe prompt to the AI chatbot.

2 / 3

KEY CONTRIBUTIONS

What PromptShield Contributes

Three core contributions to privacy-preserving conversational AI:



The prototype demonstrates prompt sanitization, explicit user control, and information governance in one unified framework.

3 / 3



Eva María Benito Sanz · 1st

Master's Student at the Technical University of Denmark

3mo · 

From course project to published research 🎓

Our first paper just got published in Sensors (MDPI):

📄 "Governing Privacy-Preserving Face Recognition in Transport Infrastructures: A Comprehensive Review"

🔗 https://lnkd.in/eKh_4RyA

Still surreal, honestly. This started as a course project with [Alba Gonzalo Primo](#) at [DTU - Technical University of Denmark](#) and somehow has become a published paper on May 1st.

I'm a Master's student in Cybersecurity with an AI and biometrics background. I naturally think about how security systems break and how to fix them. Alba's strength is mathematical modeling and AI. When we started collaborating, we kept asking questions that went beyond the technical side. And that's where it clicked: you can't evaluate facial recognition in airports or borders on security alone. Governance matters too. Accountability, privacy protection, responsible deployment.

Our paper explores that gap, it's less about better algorithms and more about the structures you actually need to deploy these systems responsibly.

Huge thanks to Alba for pushing me to think bigger, and to [Gaurav Choudhary, Ph.D.](#) and [Nicola Dragoni](#) for believing in the work and guiding us through it.

Really grateful. 🚀



sensors

an Open Access Journal by MDPI



Governing Privacy-Preserving Face Recognition in Transport Infrastructures: A Comprehensive Review

Eva María Benito Sanz; Alba Gonzalo Primo; Gaurav Choudhary; Nicola Dragoni

Sensors **2026**, Volume 26, Issue 9, 2832

Student Workshop

- A schedule will be announced on DTU Learn
- You are NOT expected to present any final solution or finalised work!
- The presentation is meant to illustrate your project, the main idea behind the project and what you plan to deliver in the end, so that you can get some immediate feedback
 - Each presentation should contain at least the following info:
 1. Type of project: technical paper or systematic literature review
 2. What the project is about (what kind of problem are you targeting? how?)
 3. Status of the project (what you have done so far)
 4. What you plan to do in the remaining time
 5. Who is going to do what
- Important: we expect everybody to present and attend the student workshop!

N.B.!

- You don't get any specific grade for the presentation at the student workshop
 - There is only a final grade that takes everything into consideration
- The student workshop is not mandatory
 - If you do not show up, you can still pass the course
- However, in course description there is a **learning objective** that states:

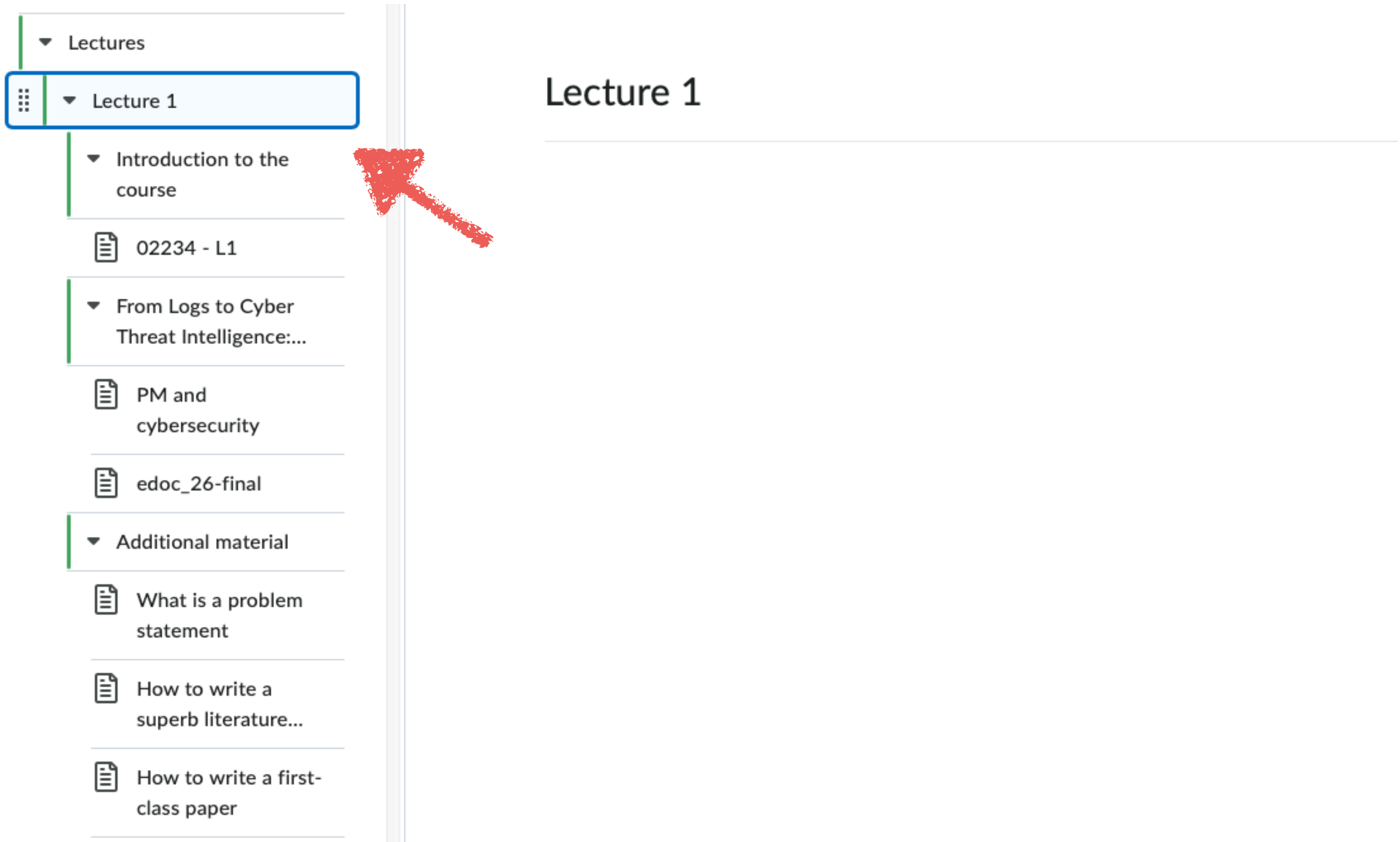
“Give a coherent oral presentation of a chosen topic or topics in system security, based on a study of relevant literature”

- If a group does not show up at the student workshop, this learning objective is **NOT** met by the members of the group
- This is taken into consideration in the final grade of the members of the group

Evaluation

- Group work but **individual evaluation**
 - Your **report should clearly state who did what**

Last but not Least... Everything is on DTU Learn



The screenshot displays the DTU Learn interface. On the left, a sidebar menu shows the course structure. The 'Lectures' section is expanded, and 'Lecture 1' is selected. Under 'Lecture 1', the following items are listed:

- Introduction to the course (highlighted with a red arrow)
- 02234 - L1
- From Logs to Cyber Threat Intelligence:...
- PM and cybersecurity
- edoc_26-final
- Additional material
 - What is a problem statement
 - How to write a superb literature...
 - How to write a first-class paper

The main content area on the right is titled 'Lecture 1'.

Questions?

After This Course...

Home

Funded Projects

Team

Publications

Invited Talks

Teaching

Thesis



NICOLA DRAGONI

Professor in Secure Pervasive Computing, PhD

Deputy Director at [DTU Compute](#)

Head of Section for [Cybersecurity Engineering](#)

DTU Compute - Department of Applied Mathematics and Computer Science
Technical University of Denmark

OFFICE

Richard Petersens Plads, building 322, room 217
2800 Kongens Lyngby, Denmark

CONTACT

Email: ndra AT dtu.dk
Phone: +45 45253731



Today's Agenda

1. Course Practicalities

- Course format
- Activity plan
- Exam
- Example of Technical Paper (Alfonso)

2. From Logs to Cyber Threat Intelligence: Mining the Behaviour of Cyber Threats

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